

Spacecraft Architecture

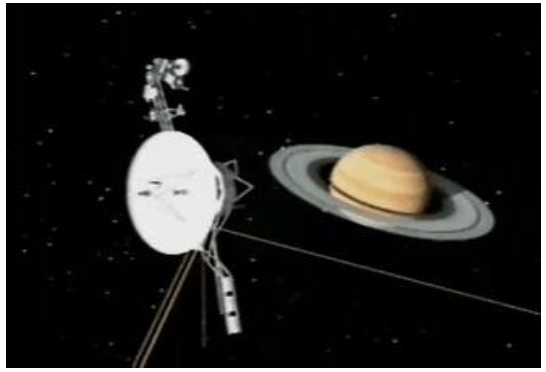
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NASA Jet Propulsion Laboratory
February 9, 2026

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Jet Propulsion Laboratory
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Introduction to Spacecraft Architectures

- Many different kinds of spacecraft
  - Explorers
  - Communications
  - Weather
  - Military
  - Cubesats



Voyager



Asteria



Iridium

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Spacecraft Subsystems

- Propulsion
  - Thrusters (main, RCS)
  - Valves
  - Fuel delivery
- Telecom
  - Data encoding
  - Amplification and transmission
  - Reception
  - Radio Science
  - Antenna system

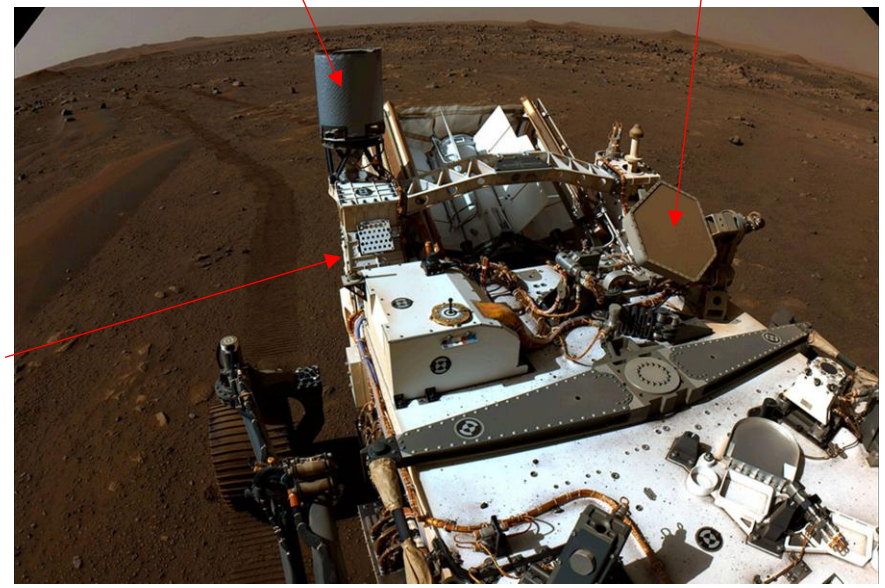


Saturn V

UHF (To orbit)

X-Band (To Earth)

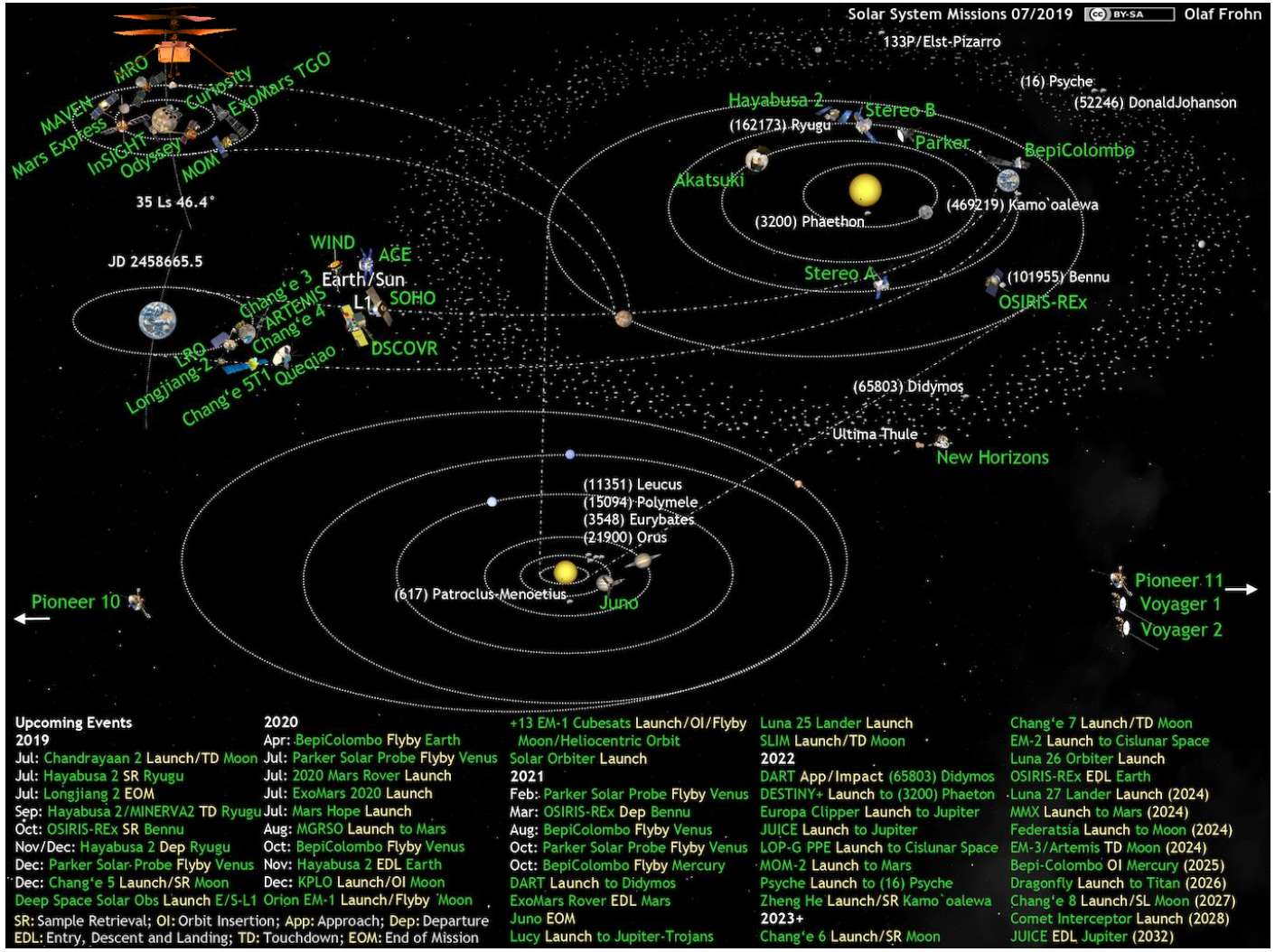
ISM (To helicopter)



Perseverance Rover

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Active Spacecraft



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Other Spacecraft Architectures

- Payloads
  - Instruments
  - Radars
  - Experiments
- Deployables
  - Aerial Vehicles
  - Rovers



Ingenuity

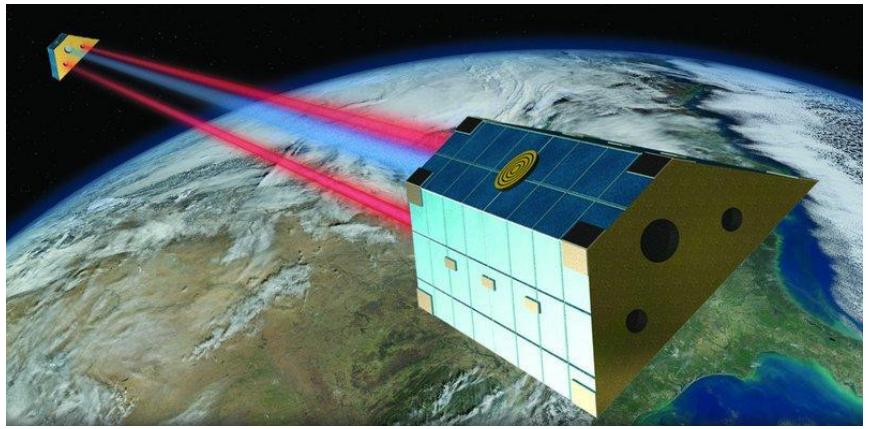
Jason 2





Cadre

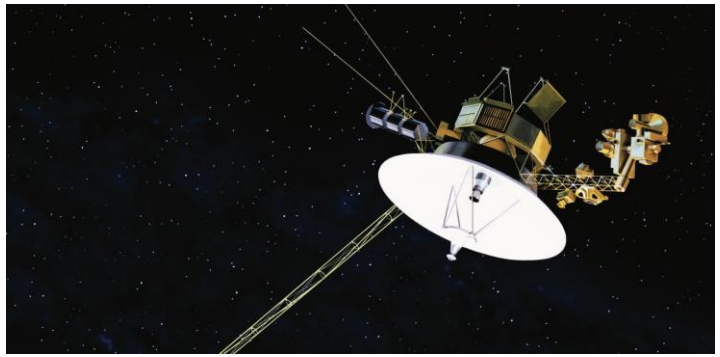
Grace Follow-On



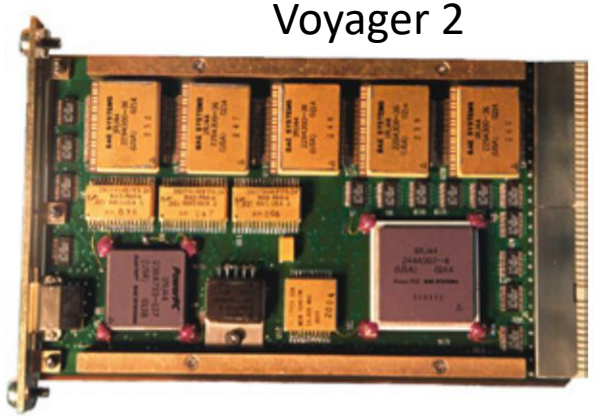
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Spacecraft Characteristics

- Characteristics of Spacecraft
  - Rugged
    - Survive for many years
    - Over-engineered
    - Old but reliable tech
  - Space-worthy
    - Radiation
    - Thermal
    - Power
  - Autonomous
    - Respond to faults
    - Guidance algorithms
    - Research on even more



Voyager 2



BAE
RAD750

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Spacecraft Subsystems

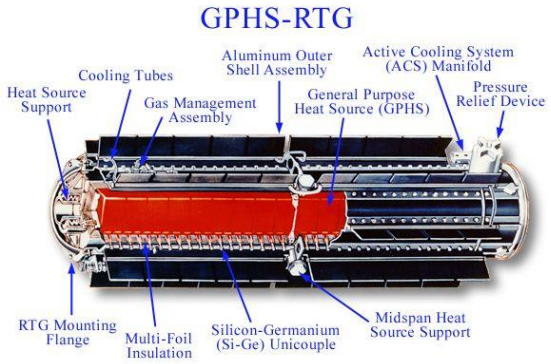
- Command and Data Handling (C&DH)
  - Computer processor(s)
  - Storage
  - Input/Output devices
    - UARTs, SpaceWire, SPI, I2C
  - Communication Busses
    - PCI, AMBA, 1553, VME
- Power
  - Power source
  - Battery system
  - Power switching

Cassini Flight Computer





Ingenuity Avionics



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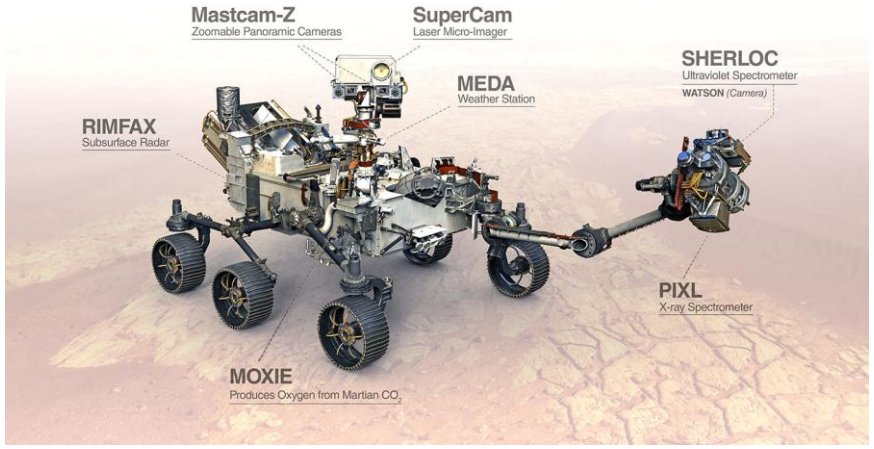
Spacecraft Subsystems

- Mechanical
  - Structure of the system
    - Rigid
    - Articulating
    - Rotating
- Payloads
  - Instruments
  - Sensors
  - Experiments
  - Imagers

Insight



Perseverance



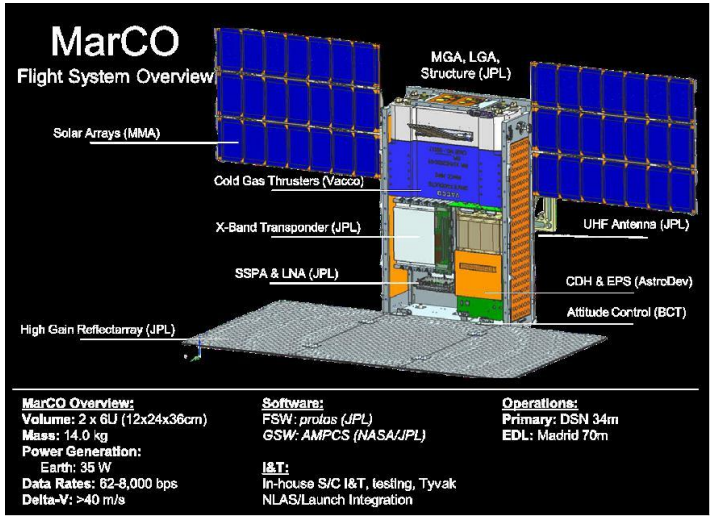
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Spacecraft Subsystems – Hardware Layering

Peripheral
(Payload, Power, Telecom, etc)

I/O
Device

Processor



MarCo Cubesat

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References (shortened via bitly.com)

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- Iridium - https://bit.ly/2gFBaUb
- Asteria - https://go.nasa.gov/30TovDQ
- Saturn V - https://bit.ly/2HDOjvZ
- Perseverance - https://go.nasa.gov/3V8zN36
- Active Spacecraft - https://bit.ly/1K61I9C
- Jason 2 - https://bit.ly/2JIDjjh
- Grace Follow-on - https://bit.ly/2HXO3XF
- Voyager 2 - https://bit.ly/2vhOh3q
- BAE RAD750 - https://bit.ly/2JFgM6T
- Cassini C&DH - https://go.nasa.gov/2wlbbtN
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- MarCo - https://go.nasa.gov/2WtrAuN

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